Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Projects

Rayfloat | *C++, OpenMP* | GitHub | Writeup — December 2025

- Architected a physically-based CPU renderer from scratch, implementing Lambertian, Metal, Dielectric, and DiffuseLight materials with an **iterative light transport model**, eliminating **recursion-induced stack pressure** and improving **Branch Target Buffer (BTB) predictability**.
- Diagnosed and eliminated a **41% performance penalty** from **cache-coherence contention** in `std::mt19937` by implementing a **thread-local XorShift32 pRNG**, restoring **near-linear multi-core scaling**.
- Reduced ray-scene intersection complexity from **O(n) to O(log n)** by designing a **Bounding Volume Hierarchy (BVH)** with **median-split construction** and optimized slab AABB tests, enabling scenes with 64,000+ primitives.
- Optimized **dielectric rendering** using **Schlick's Approximation** to numerically estimate the **Fresnel coefficient**, determining the probability of reflection vs refraction based on incident angles.

TouchGrass ChessEngine | *Python* | GitHub — October 2025

- Built a fully rules-accurate chess backend with legal move generation, state transitions, threefold repetition detection, and a complete undo/redo stack
- Integrated Zobrist hashing for efficient board state encoding, enabling fast repetition tracking and transposition lookups
- Implemented a modular alpha-beta search atop the backend's move execution system, supporting promotions, castling, and en passant
- Designed an extensible API exposing board state, legal moves, move execution, and termination detection for engine or UI clients

amine | *Python, Flask, PyAutoGUI, SQLite* | GitHub | ProductHunt — August 2024 – October 2024

- Reached #10 on ProductHunt with 450+ downloads by building a multi-threaded distraction-blocking productivity app
- Implemented system-level mouse boundary enforcement, keyboard interception, and full-screen tracking to prevent app switching
- Designed an analytics dashboard using Matplotlib, Seaborn, and Plotly with session insights and CSV export
- Engineered SQLite-backed session management, desktop notifications, and input control, gaining 170+ GitHub stars

yars | *Python, requests, urllib3* | GitHub — October 2024

- Bypassed rate limits and anti-bot filters using automated retries, exponential backoff, proxy rotation, and randomized user agents
- Implemented global and subreddit-targeted search with pagination handling, returning structured metadata-rich results
- Built a recursive comment extraction system parsing nested Reddit threads into hierarchical JSON trees
- Developed user activity and subreddit feed scrapers supporting multiple sorting modes with controlled request scheduling

Skills

Languages: Python, C/C++, SQL
Frameworks and Libraries: OpenMP, Raylib, PyAutoGUI, pandas, NumPy, Matplotlib
Developer Tools: Git, Intel VTune Profiler, SQLite
Competitive Programming: Codeforces Pupil (1338)